THE DIAGONAL FIBRE IDENTITY FOR LOCAL RING DIMENSIONS ALONG THE GROUPOID S-PROJECTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-geometry.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for stacks-geometry.tex (ge-occupied.json, 2026-10-14); 207 route=geometry / chapter-list field hits in focused-seeds are alias noise (route name and alphabetical chapter enumerations), excluded per the ladder alias-blind-spot caveat.

Source reference. Original reference 00ON: sources/stacks-chow/tags.txt, source lines 901–901. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be a locally Noetherian algebraic stack. Let $U \to \mathcal{X}$ be a smooth morphism and let $u \in U$. Then*

$$\dim(\mathcal{O}_{U,\overline{u}}) - \dim(\mathcal{O}_{R_u,e(\overline{u})}) = 2\dim(\mathcal{O}_{U,\overline{u}}) - \dim(\mathcal{O}_{R,e(\overline{u})})$$

Here $R = U \times_{\mathcal{X}} U$ with projections $s, t : R \to U$ and diagonal $e : U \to R$ and R_u is the fibre of $s : R \to U$ over u.

Proof. This is true because $s : \mathcal{O}_{U,\overline{u}} \to \mathcal{O}_{R,e(\overline{u})}$ is a flat local homomorphism of Noetherian local rings and hence

$$\dim(\mathcal{O}_{R,e(\overline{u})}) = \dim(\mathcal{O}_{U,\overline{u}}) + \dim(\mathcal{O}_{R_u,e(\overline{u})})$$

by Algebra, Lemma 00ON. □

Date: stacks-project:stacks-geometry.tex:local-snapshot-20261012; GNU FDL 1.2 or later.